Date: 6 Mar 2026

Project Name: UCab - Modern Ride-Sharing Platform

Maximum Marks: 2 Marks

Ideation Phase

Define the Problem Statements

Customer Problem Statement Template: Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

Reference: https://miro.com/templates/customer-problem-statement/

Problem Statements Table

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A daily commuter / office-goer	book a reliable, safe, and affordable cab to reach my workplace on time	I face long waiting times, frequent ride cancellations, and unpredictable surge pricing	existing apps have poor driver-matching algorithms and drivers cancel if the drop location isn't favorable to them	Frustrated, anxious about being late, and helpless during peak hours.

PS-2	A full-time cab driver	earn a steady and fair income by completing daily rides efficiently	I spend too much time driving empty (dry runs) and lose a huge chunk of my earnings to platform commissions	the current platforms lack transparent fare distribution and do not optimize nearby ride allocations effectively	Underpaid, exhausted, and unmotivated to accept long-distance rides.
PS-3	A late-night traveler / female commuter	travel safely to my destination with complete peace of mind	I feel unsafe and cannot accurately share my live location with my family in real-time	current apps often have laggy GPS tracking and lack transparent driver verification details	Vulnerable, anxious, and hesitant to book cabs at night.

Proposed Solution (UCab):

To address the above challenges, we are developing **UCab**, a modern, full-stack ride-sharing platform built on the MERN stack. Our solution focuses on three core pillars:

- **Instant & Reliable Matching (Solves PS-1):** Using MongoDB's geospatial queries, UCab connects riders with the absolute nearest drivers, minimizing wait times and reducing cancellation rates.
- **Transparent Earnings & Payments (Solves PS-2):** UCab integrates a direct QR-code-based UPI payment system, ensuring drivers get their fair share instantly without hidden platform deductions.
- **Real-Time Precision Tracking (Solves PS-3):** Implemented using Socket.io (WebSockets), UCab provides zero-latency, live location tracking of the cab, allowing users to feel safe and easily monitor their journey second-by-second.